**SIMON OKOMO ALOO**

IELTS Overall: 7.0 | LinkedIn: www.linkedin.com/in/simonaloo | Google Scholar: simon okomo aloo - Google Scholar

## EDUCATION

**2025-Date:** Doctor of Philosophy, Monash University, Australia (Ongoing)

Research area: Tracing lipid metabolism to uncover lipid mediated cardiometabolic disorder mechanisms

**2022-2024**: Doctor of Philosophy (Food Science and Biotechnology) Kangwon National University, Korea

Thesis: Ameliorative effects of Pediococcus acidilactici KCTC 15831BP fermented hempseed on obesity and obesity-induced Alzheimer's disease: insights from microbiome and metabolomics

GPA: 4.32/4.5 (96%)

**2020-2022:** Master of Science (Food Science and Biotechnology), Kangwon National University, Korea)

Thesis: *Anti-obesity effect of alfalfa sprout fermented with Pediococcus acidilactici in Caenorhabditis elegans model.*

GPA: 4.49/4.5 (99%)

**2014-2018:** Bachelor of Science (Food Science and Nutrition), Karatina University, Kenya

Thesis: *Effects of processing on the biochemical profile and proximate composition of tempeh made from lablab bean.* GPA: 3.82/4 (96%). *Category: First Class Honors*

## RESEARCH ACHIEVEMENT SUMMERY

**Publication history:** Total Publications -28 | Total first author articles - 19 | Total original articles -18 | Total review articles- 8 | Book chapters- 2 | Q1 publications -24 | Q2 publications - 2 | Articles in top 10% journals - 8

**ResearchGate:** 715 citations | h-index: 15 | Research Interest Score 482.0 | Google Scholar: 790 citations | i10-index

**Previous Research:** Developed probiotic-fermented plant-based foods with therapeutic potential against obesity and obesity-associated neurodegeneration, using C. elegans and mouse models, and integrating microbiome and metabolomics analyses to elucidate underlying mechanisms and functional bioactivity.

**Current Research:** Developing and applying stable isotope lipid tracer approaches in individualized cultured human cells to uncover genetic regulation of lipid metabolism, aiming to identify genetic drivers of lipid dysregulation in cardiometabolic diseases.

## OUTSTANDING AWARDS

**2022:** Outstanding Academic Achievement Award, Kangwon National University

Top master's student for the academic year: Awarded on 28th February 2022.

**2021:** Outstanding Academic Achievement Award, National Institute for International Education (Korea)

Top master's candidate among Korean Government Scholars: Award presented by the Deputy Prime Minister, Republic of Korea

**2020:** Excellence Award, Korea Youth Work Agency

3rd place essay contest: Awarded by CEO of Korea Youth Work Agency

**2020:** Best Speaker's Award, Global Youth Leader Exchange Program

Recognized as an outstanding speech during a TED contest 'Covid 19 and Life changes'

**2018:** Excellence Award, Asia-Pacific African Academy (APACA)

Recognized as an outstanding participant during 2018 Training Program for Young Upcoming African Agricultural Experts

## SCHOLARSHIP AWARDS

**2025-date:** Monash Graduate Scholarship (MGS) award, Monash University

**2025:** Australian Government Research Training Program (AGRTP) Scholarship, University of Wollongong

**2022-2024:** Bring Excellent Students to Kangwon National University (BEST-KNU) scholarship

**2022-2024:** Brain Korea 21 (BK21) scholarship, Kangwon National University

**2019-2022:** Korean Government Graduate Scholarship (KGSP), Kangwon National University

**2018:** Global Korea Scholarship for Undergraduate students, Asia-Pacific Africa Co-operation Academy (apaca).

## INTERNATION CONFERENCE: ORAL PRESENTATIONS

Aloo, S. (2024). *Pediococcus acidilactici–fermented hempseed alleviates obesity and improves Alzheimer's disease–like symptoms via oxido-inflammatory pathways* [Oral presentation]. Korean Food Science and Nutrition (KFN) Society 2024 International Annual Meeting, **Jeju, South Korea.**

Aloo, S. (2023). *Protective effect of fermented hempseed extracts against obesity and obesity-induced oxidative stress and neurodegenerative disorders* [Oral presentation]. 16th International Scientific Conference on Probiotics, Prebiotics, Gut Microbiota and Health, **Bratislava, Slovakia**.

Aloo, S. (2023). *Combination of elicited germination and fermentation improves metabolite profile and anti-obesity properties of alfalfa seed* [Oral presentation]. Korean Society of Food Science and Technology (KoSFoST) International Symposium and Annual Meeting, **Busan, Korea.**

## TRAINING

**2023:** *Kangwon National University*

Topic: Ethical guidelines for the use of experimental animals and related skills.

**2021:** *Netmission Academy.Asia, Hong Kong*

Topic: Internet governance and human rights online

**2020:** *International Youth Leader Exchange Program*

Topic: World economy post-COVID-19 era.

**2018:** *Asia Pacific-African Cooperation Academy*

Topic: Emerging technologies in agriculture.

## PROFESSIONAL EXPERIENCE

**2022-2024: Food Technologist - Future Biotech Ltd, Republic of Korea**

**Responsibilities**

- Conducted industry collaborative research on probiotic and fermented food development.
- Participated in pilot-scale fermentation and process standardization.
- Developed and evaluated probiotic formulations for metabolic health benefits.

**2017: Food Science Intern - Kenya Bureau of Standards (KEBS), Kenya**

**Responsibilities**

- Assisted in food quality analysis and product standardization activities.
- Conducted laboratory testing for physicochemical properties of foods.

- Supported inspection and certification processes for locally manufactured food products.

**2016: Clinical Nutrition Intern - Homa-Bay Referral Hospital, Kenya**

- Assessed patients' nutritional status and developed individualized diet plans.
- Assisted in management of diet-related conditions (diabetes, hypertension, malnutrition).
- Participated in hospital feeding program evaluation and dietary monitoring.

## LEADERSHIP AND COMMUNITY ENGANGEMENTS

**March 2026-Date:** Monash University, Science, Technology, and Medicine (STM) Student mentor

- Provide peer mentorship to honor students by offering guidance on the research experience, fostering a supportive and inclusive environment, and helping mentees navigate university resources.

**2025-Date:** Chairperson, Alliance for Cardiometabolic Health & Research–Kenya (ACMHR-K).

- Provide leadership for a national non-profit focused on cardiometabolic disease prevention through interdisciplinary research, training, and policy advocacy (https://acmhr-k.org/).

**2022-2024:** Board Clerk & Campus Ministry Coordinator, Harvest Time Intl. Church, Korea.

- Supported needy student through campus ministry fellowship at both Hallym University and Kangwon National University campuses

**2020-2021:** Kangwon National University-Team Leader, International University Promotion Committee

- Led an international team in promoting Kangwon National University to prospective students and global partners, supporting international recruitment, visibility, and institutional outreach.

**2020-2021:** African Youth Entrepreneurship Academy, Tutor in the sector of agriculture

- Mentored and trained young entrepreneurs and Agri-professionals across Africa, building capacity in agribusiness innovation, value addition, and sustainable agricultural entrepreneurship

## SKILLS AND KNOWLEDGE

- Scientific Writing: Experienced in synthesizing research ideas and findings into high-quality manuscripts for international publication.
- Animal Studies (Mouse Models): Experienced in designing and conducting mouse experiments, including oral gavage and the establishment of high-fat diet–induced models.
- Caenorhabditis elegans Research: proficient in culturing and maintaining C. elegans, and in developing high-glucose–induced models for metabolic studies.
- Cell Culture Techniques: Average skills in cell line maintenance, aseptic handling, and experimental design.

**First author articles**


1. Aloo, Simon Okomo, Ramachandran Chelliah, Fred Kwame Ofosu, Ye-Won Lee, Ye Eun Cho, Se Jin Park, and Deog Hwan Oh. "Pediococcus acidilactici KCTC 15831BP-fermented industrial hempseed (Cannabis sativa L.) supplementation corrects metabolite and gut microbiota dysbiosis, potentially mitigating Alzheimer's disease-like symptoms induced by obesity in high-fat diet-fed mice." Food & function (2025): 40878211.
2. Aloo, Simon Okomo, SeonJu Park, Timilehin Martins Oyinloye, Ye-Won Lee, Ye Eun Cho, Se Jin Park, and Deog-Hwan Oh. "Effects of liquid state vs. solid state lactic fermentation on drying, nutritional composition, phytochemical profile, and in vitro neuro-related bioactivities of Cheungsam industrial hempseed (Korean breed)." Food Bioscience 63 (2025): 105708.

3. Aloo, Simon Okomo, SeonJu Park, Ye-Jin Jeong, Tuaumelsan Shumye Gebre, and Deog-Hwan Oh. "Characterization of different hempseed fractions after dehulling and defatting: Chemical composition and functional properties." Journal of Food Measurement and Characterization 18, no. 9 (2024): 7868-7883.
4. Aloo, Okomo Simon, Feyera Gobena Gemechu, Hyun-Ji Oh, Emily Cherono Kilel, Ramachandran Chelliah, Girma Gonfa, and Deog-Hwan Oh. "Harnessing fermentation for sustainable beverage production: A tool for improving the nutritional quality of coffee bean and valorizing coffee byproducts." *Biocatalysis and Agricultural Biotechnology* 59 (2024): 103263.
5. Aloo, Okomo Simon, Dong-Gyu Kim, Selvakumar Vijayalakshmi, Denish Obiero Aloo, Charles O. Ochola, and Deog-Hwan Oh. "Polyphenol constituents and impacts of fermented teas (Camellia sinensis) in human wellness." Food Bioscience 60 (2024): 104389.
6. Aloo, Simon Okomo, Godfrey Mwiti, Louise Wanjiku Ngugi, and Deog-Hwan Oh. "Uncovering the secrets of industrial hemp in food and nutrition: The trends, challenges, and new-age perspectives." Critical Reviews in Food Science and Nutrition 64, no. 15 (2024): 5093-5112.
7. Aloo, Simon Okomo, SeonJu Park, Timilehin Martins Oyinloye, and Deog-Hwan Oh. "Rheological properties, biochemical changes, and potential health benefits of dehulled and defatted industrial hempseeds after fermentation." Food Chemistry 439 (2024): 138086.
8. Aloo, Simon Okomo, Kaliyan Barathikannan, and Deog-Hwan Oh. "Polyphenol-rich fermented hempseed ethanol extracts improve obesity, oxidative stress, and neural health in high-glucose diet-induced Caenorhabditis elegans." Food Chemistry: X 21 (2024): 101233.
9. Aloo, Simon Okomo, SeonJu Park, and Deog-Hwan Oh. "Impacts of germination and lactic acid bacteria fermentation on anti-nutrients, bioactive compounds, and selected functional properties of industrial hempseed (Cannabis sativa L.)." Food Chemistry 428 (2023): 136722.
10. Aloo, Simon Okomo, Fred Kwame Ofosu, and Deog-Hwan Oh. "Elicitation: a new perspective into plant chemo-diversity and functional property." Critical reviews in food science and nutrition 63, no. 20 (2023): 4522-4540.
11. Aloo, Simon Okomo, Kaliyan Barathikannan, Fred Kwame Ofosu, and Deog-Hwan Oh. "Fermentation of ascorbic acid-elicited alfalfa sprouts further enhances their metabolite profile, antioxidant, and anti-obesity effects." Food Bioscience 54 (2023): 102871.
12. Aloo, Simon Okomo, Fred Kwame Ofosu, Mary Njeri Muchiri, Selvakumar Vijayalakshmi, Choi-Geun Pyo, and Deog-Hwan Oh. "In Vitro Bioactivities of Commonly Consumed Cereal, Vegetable, and Legume Seeds as Related to Their Bioactive Components: An Untargeted Metabolomics Approach Using UHPLC–QTOF-MS2." Antioxidants 12, no. 8 (2023): 1501.
13. Aloo, Simon-Okomo, Fred Kwame Ofosu, Nam-Hyeon Kim, Sheila M. Kilonzi, and Deog-Hwan Oh. "Insights on dietary polyphenols as agents against metabolic disorders: obesity as a target disease." Antioxidants 12, no. 2 (2023): 416.
14. Aloo, Simon Okomo, Fred Ofosu Kwame, and Deog-Hwan Oh. "Identification of possible bioactive compounds and a comparative study on in vitro biological properties of whole hemp seed and stem." Food Bioscience 51 (2023): 102329.
15. Aloo, Simon Okomo, and Deog-Hwan Oh. "The functional interplay between gut microbiota, protein hydrolysates/bioactive peptides, and obesity: a critical review on the study advances." Antioxidants 11, no. 2 (2022): 333.
16. Aloo, Simon-Okomo, Fred-Kwame Ofosu, and Deog-Hwan Oh. "Effect of germination on alfalfa and buckwheat: phytochemical profiling by UHPLC-ESI-QTOF-MS/MS, bioactive compounds, and in-vitro studies of their diabetes and obesity-related functions." Antioxidants 10, no. 10 (2021): 1613.**.**
17. Aloo, Simon Okomo, Fred Kwame Ofosu, Sheila M. Kilonzi, Umair Shabbir, and Deog Hwan Oh. "Edible plant sprouts: Health benefits, trends, and opportunities for novel exploration." Nutrients 13, no. 8 (2021): 2882.
18. Aloo, Simon-Okomo, Fred-Kwame Ofosu, Eric-Banan-Mwine Daliri, and Deog-Hwan Oh. "UHPLC-ESI-QTOF-MS/MS metabolite profiling of the antioxidant and antidiabetic activities of red cabbage and broccoli seeds and sprouts." Antioxidants 10, no. 6 (2021): 852.

**Co-author articles**

19. Ofosu, Fred Kwame, Fazle Elahi, Eric Banan-Mwine Daliri, **Simon Okomo Aloo**, Ramachandran Chelliah, Sang-Ik Han, and Deog-Hwan Oh. "Fermented sorghum improves type 2 diabetes remission by modulating gut microbiota and their related metabolites in high fat diet-streptozotocin induced diabetic mice." *Journal of Functional Foods* 107 (2023): 105666.
20. Shabbir, Umair, Akanksha Tyagi, Fazle Elahi, **Simon Okomo Aloo,** and Deog-Hwan Oh. "The potential role of polyphenols in oxidative stress and inflammation induced by gut microbiota in alzheimer's disease." *Antioxidants* 10, no. 9 (2021): 1370.

21. Gebre, Tuaumelsan Shumye, Shimelis Admassu Emire, **Simon Okomo Aloo,** Ramachandran Chelliah, Selvakumar Vijayalakshmi, and Deog Hwan Oh. "Unveiling the potential of African fermented cereal-based beverages: Probiotics, functional drinks, health benefits and bioactive components." *Food Research International* 191 (2024): 114656.
22. Gebre, Tuaumelsan Shumye, Shimelis Admassu Emire, Ramachandran Chelliah, **Simon Okomo Aloo,** and Deog-Hwan Oh. "Isolation, functional activity, and safety of probiotics from Ethiopian traditional cereal-based fermented beverage,"Borde"." *Lwt* 184 (2023): 115076.
23. Vijayalakshmi, Selvakumar, Dae-Sang Yoo, Dong-Gyu Kim, Ramachandran Chelliah, Kaliyan Barathikannan, **Simon-Okomo Aloo,** Akanksha Tyagi et al. "Fermented Perilla frutescens leaves and their untargeted metabolomics by UHPLC-QTOF-MS reveal anticancer and immunomodulatory effects." *Food Bioscience* 56 (2023): 103065.
24. Gebre, Tuaumelsan Shumye, Shimelis Admassu Emire, Kaliyan Barathikannan, **Simon Okomo Aloo,** Ramachandran Chelliah, and Deog Hawn Oh. "UHPLC-Q-TOF-MS/MS identification of metabolites in cereal beverage 'borde'and its anti-obesity efficacy in Caenorhabditis elegans model." *Process Biochemistry* 146 (2024): 325-336.
25. Vijayalakshmi, Selvakumar, Jong-Rai Kim, Ramachandran Chelliah, Kaliyan Barathikannan, Akanksha Tyagi, **Simon-Okomo Aloo**, Xiuqin Chen, Pianpian Yan, Lingyue Shan, and Deog-Hwan Oh. "Encapsulating potential and functional properties of exopolysaccharide from Limosilactobacillus reuteri KCTC 14626BP isolated from human breast milk." *International Journal of Biological Macromolecules* 253 (2023): 127330.
26. Yan, Pianpian, **Simon Okomo Aloo**, Lingyue Shan, Akanksha Tyagi, and Deog Hwan Oh. "Effects of slightly acidic electrolyzed water on safety, antioxidant activity, and metabolite profile of buckwheat sprouts (Fagopyrum spp.)." *Journal of Cereal Science* 118 (2024): 103969.

Book Chapters

1. Aloo, Simon Okomo, Denish Obiero Aloo, Selvakumar Vijayalakshmi, and Deog-Hwan Oh. "Nucleic Acid-Based Rapid Detection of Plant-Associated Cyanobacteria." In *Plant Microbiome Engineering*, pp. 151-160. New York, NY: Springer US, 2024.
2. Aloo, Simon Okomo, Denish Obiero Aloo, Selvakumar Vijayalakshmi, and Deog-Hwan Oh. "Identification of Toxic Blooms of Cyanobacteria in Marine Water Habitat." In *Protocols for Cyanobacteria Sampling and Detection of Cyanotoxin*, pp. 17-25. Singapore: Springer Nature Singapore, 2023.

**Profile summary**

Simon Aloo earned a Bachelor of Science with First Class Honours in Food Science and Nutrition from Karatina University. As an undergraduate, he was selected for a competitive international training program titled the "Korean Government Program for Young Upcoming African Agricultural Experts." Hosted by the Asia Pacific–Africa Cooperation Academy in South Korea, the program brought together 20 outstanding undergraduate students from across Africa for a capacity-building experience in agricultural innovation. At the conclusion of the program, he was honored with the Excellence Award, a distinction granted to only one outstanding participant in recognition of exceptional performance. In 2019, he was awarded the prestigious GKS-G scholarship. He studied the Korean language at Konyang University and then proceeded to graduate studies at Kangwon National University (KNU), South Korea. There, he pursued a master's degree in food science and biotechnology and subsequently earned his first PhD from the same university after two years. He was awarded Outstanding Research Achievement by KNU for both his master's and PhD programs. During this period, he also received numerous prestigious awards, including the NIIED Award of Excellence (2020), the BEST-KNU scholarship, and the Brain Korea (BK21+) Research Fellowship (2021), all reserved for exceptional scholars. After completing his first doctorate, he was awarded two additional prestigious scholarships: the Australian Research Training Scholarship at the University of Wollongong and the Monash Graduate Scholarship. He is now affiliated with Monash University's School of Translational Medicine at the Baker Heart and Diabetes Institute–Alfred Hospital, where his research explores the links between lipid

dysregulation and chronic diseases. He is also the founder of ACMHR-K, a registered NGO that promotes cardiometabolic research and health education in Kenya. He takes pride in assisting young Africans who are interested in studying abroad by helping them navigate scholarship applications and make informed career choices.